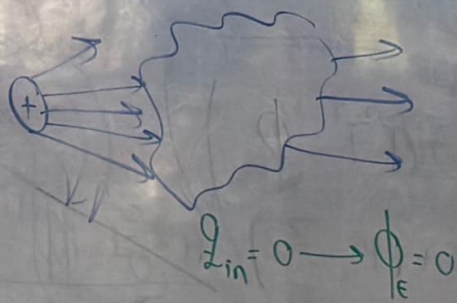
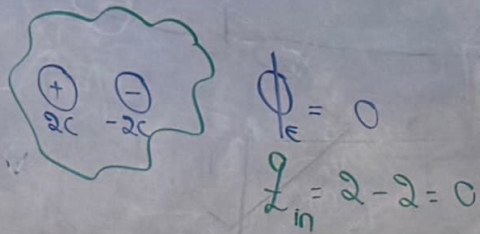
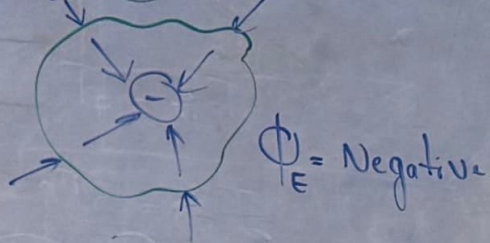
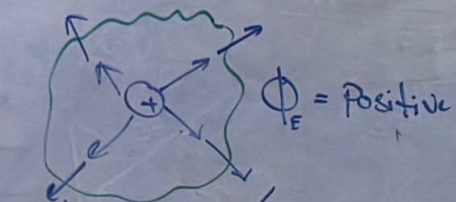


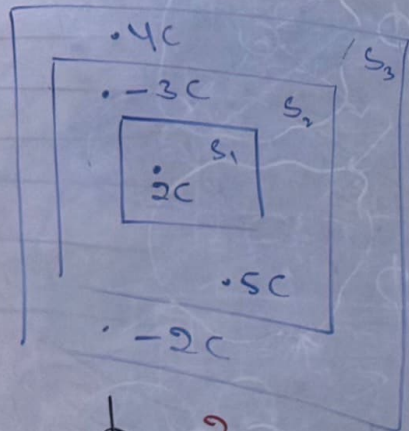
Gauss's law

it is used to calculate the electric flux through a closed surface
closed surface: Any surface that divides a space into inside and outside regions.

examples: Cube - Sphere



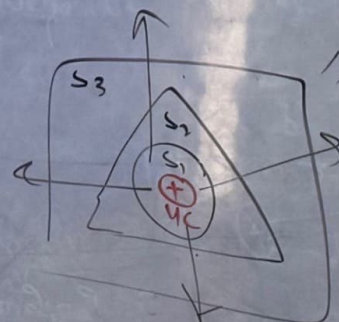
- ① entering lines are more
- ② leaving lines are more
- ③ entering = leaving



$$\phi_{s_1} = \frac{2}{\epsilon_0}$$

$$\phi_{s_2} = \frac{5+2-3}{\epsilon_0} = \frac{4}{\epsilon_0}$$

$$\phi_{s_3} = \frac{4+5+2-2-3}{\epsilon_0} = \frac{6}{\epsilon_0}$$



Gauss's law

The total electric flux through any closed surface is

$$\phi_E = \frac{q_{in}}{\epsilon_0}$$

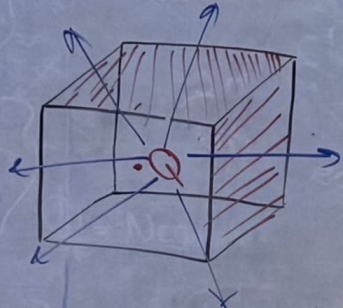
for closed surface

electric flux depends only on inside charge

" is independent of shape of a closed surface
radius

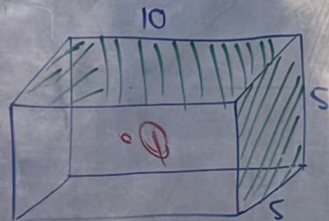
Q16

A charge of $Q = 26.55 \times 10^{-12} \text{ C}$ is placed at a center of a cube.
Find the electric flux through shaded areas.



$$\Phi = \frac{2q_{\text{in}}}{6\epsilon_0} = \frac{q_{\text{in}}}{3\epsilon_0}$$

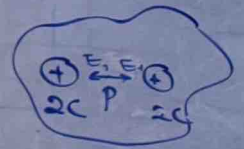
$$\Phi_E = \frac{2}{6} \frac{q_{\text{in}}}{\epsilon_0} = \frac{q_{\text{in}}}{3\epsilon_0} = \frac{26.55 \times 10^{-12}}{3 \times 8.85 \times 10^{-12}} = 1 \text{ N.m}^2/\text{C}$$



$$\Phi_E = \frac{1.5 q_{\text{in}}}{5\epsilon_0}$$

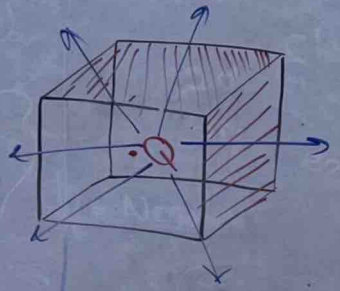
Q16

A charge of $Q = 26.55 \times 10^{-12} \text{ C}$ is placed at a center of a cube. Find the electric flux through shaded areas.

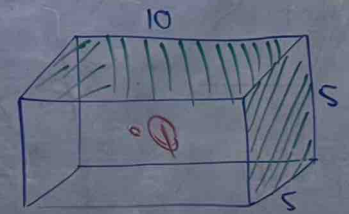


$$E = E_1 - E_2 = 0$$

$$\phi = q/\epsilon$$



$$\phi_E = \frac{2}{6} \frac{q_{in}}{\epsilon} = \frac{q_{in}}{3\epsilon} = \frac{26.55 \times 10^{-12}}{3 \times 8.85 \times 10^{-12}} = 1 \text{ N}\cdot\text{m}^2/\text{C}$$



$$\phi_E = \frac{1.5 q_{in}}{5\epsilon}$$

10 N·m²/C